

Plasticity in evolutionary games

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Abstract

The ability to respond appropriately to environmental cues is fundamental to the success of all forms of life. However, previous theoretical studies of the evolution of plasticity make very diverse assumptions, so the conditions under which plasticity can emerge in evolving populations are unclear when fitness is frequency-dependent. In order to contribute to addressing this question, we study the effect of adding plastic types to symmetric evolutionary games. Since frequency dependence induces an evolutionary change in the environment of players, one might expect that plastic individuals who can adapt their phenotypes to the environment would have a fitness advantage over simpler purely genetically determined phenotypes. In our model, plastic individuals can detect the type of their opponent before an interaction and condition their action on it. Even though this ability might appear to be an outstanding advantage, we find that it cannot guarantee invasion of pure types by plastic types in all games as long as plasticity is costly. We classify games according to whether plasticity can or cannot invade a population of pure types and become the evolutionarily stable strategy. In games where the standard replicator dynamics converge to a pure state, costly plasticity cannot invade an equilibrium population. One can guarantee co-existence of plasticity with pure types in games with mixed equilibria, mirroring the result of frequency-independent models that plasticity thrives when there is variance in fitness. Costly plasticity can however be locally stable in many games, but the way to achieve stability is not to play the best response to any possible encountered type. Rather, part of the stability success of plastic types is based on establishing Pareto-efficiency as residents. Zero-sum games always allow for the global stability of plastic types. In model extensions where the plastic type makes errors in recognizing other types, simulations suggest that [...] for an evolving plastic response to invade pure populations by random mutations. Overall, this study offers a more principled way of thinking about the evolutionary emergence of plasticity in social scenarios. Our study also helps demonstrate that such an emergence is strongly dependent on the type of game individuals are faced with. Finally, optimal plastic responses arise only under a very narrow set of conditions.

Keywords: reaction norms; frequency dependence; natural selection; evolutionary games; variance in fitness.

1 Introduction

Most phenotypes are plastic; indeed, traits are in general only expressed in reaction to an environmental cue. The immune system is programmed to detect pathogens, quorum sensing in bacteria conditions gene expression on cell density, plant growth depends on external light via photosynthesis (Diggle et al., 2007; Bergstrom and Dugatkin, 2016). Being plastic thus seemingly provides a selective advantage to biological organisms. The conditions favoring plasticity evolution have been studied in detail in evolutionary biology, and a general conclusion that can be drawn from classical theory is that a varying environment is required for plastic traits to provide a fitness benefit (Gomulkiewicz and Kirkpatrick, 1992; Gavrillets and Scheiner, 1993a,b). However, most of these classical models were developed under the assumption that fitness depends on the environment but not on the phenotype of other organisms in the population, i.e., in the absence of frequency dependence. It is an open question under which

conditions plasticity emerges in frequency-dependent scenarios. In particular, it is unclear how the idea that varying environments favor plasticity evolution can be generalized to cases with frequency-dependent selection.

Because the key to plasticity evolution is variance in fitness, one actually does not require that the rules of the game change over evolutionary time (Dridi and Lehmann, 2014; Weitz et al., 2016; Hilbe et al., 2018) for frequency dependence to generate variance in fitness (Graves and Weinreich, 2017). Indeed, in frequency-dependent selection, other types make up part of the environment, and as long as there is some level of mixing between types (e.g., there is not full assortment between individuals of similar phenotype), any individual is going to interact with different types and hence is going to experience variance in fitness. These observations suggest that even in the absence of changes in the structure of the evolutionary game determining individuals' fitness, there is potentially a selection pressure in favor of plasticity. In order to further examine this question, we must determine what it means to have a plastic phenotype in a game theoretical context.

Plasticity in social evolution has been assumed — without investigating its evolutionary emergence — in order to address many different questions, from studies of cooperation in the iterated prisoner's dilemma (Axelrod and Hamilton, 1981; Nowak, 2006; Press and Dyson, 2012; Adami and Hintze, 2013) to investigations of signalling (Zahavi, 1975; Grafen, 1990; Maynard-Smith and Harper, 2004), and learning (McElreath and Boyd, 2007; Dridi and Lehmann, 2015, 2016), so we lack a unified view on plasticity in social evolution. These studies have provided interesting insights into our understanding of how natural selection shapes complex strategies for repeated interactions. However, complex strategies that allow an individual to condition behavior on the environment, on opponents' behavior, or on memory of past events are only possible if, in the first place, individuals possess the ability to express plastic social phenotypes. Previous work seemingly does not address the question of the evolutionary emergence of social plasticity, so it remains unclear what is the main advantage of plasticity in social evolution.

In this paper, we adopt one of the simplest implementations of plasticity that we can think of, namely we assume that plastic individuals can detect the type of their opponent before an interaction takes place, and can condition their action on the detected type. At first, this might seem to be a considerable advantage to plastic types, but we will see below that even the smallest cost impedes plasticity to dominate other strategies in all circumstances. Another potential concern is that the genetic and molecular machinery necessary to perform a combination of strategy detection and appropriate response might be complex to evolve even for the most basic forms of plasticity. A perfect response to existing types is unlikely to emerge out of a background of pure genetic determination. Indeed, previous research suggests that such perfect responses might be very difficult to evolve (McNamara et al., 1999; André and Day, 2007). For this reason, we allow our plastic types to adopt any possible response to their opponents, in contrast to a previous work on the topic (Banerjee and Weibull, 1995), where these authors have assumed that plastic types always play a best response to pure types. Here, we rather study how the evolutionary success of plastic types depends on their response to pure types. Moreover, the perceptual system allowing one to infer others' strategies might at first also be defective if it evolves from a state where there was no perceptual system of this kind in the ancestral population. We capture such imperfections by imposing a fitness cost on the expression of the plastic phenotype.

In the following, we define a model that makes our assumptions more precise, and analyze the evolutionary performance of plastic types when pitted against individuals who can only express a fixed pure strategy in a normal-form game. We start by giving a special focus to 2×2 games and analyze the replicator dynamics for four standard games of cooperation: the Prisoner's dilemma, the Stag-hunt game, the Snowdrift game, and a resource consumption game. We then generalize those results and provide classes of games where plastic types can or cannot be globally or locally stable under the standard replicator dynamics.

2 Model

We consider the standard model of evolutionary game theory (Taylor and Jonker, 1978) of a well-mixed population in which players are matched randomly in pairs to play a 2-player n -action game, which is called throughout the pure-type game. We denote the set of actions by $\mathcal{A}$. The population consists of $n + 1$ types: the first $n = |\mathcal{A}|$ types are called pure types, and correspond to each pure action while the $(n + 1)$ -th type can detect others' type before choosing an action. This plastic type, denoted p , has a strategy described by the vector $\mathbf{z} = (\mathbf{z}_1, \dots, \mathbf{z}_n, \mathbf{z}_{n+1})$, where $\mathbf{z}_i = (z_{i1}, \dots, z_{in}) \in \Delta\mathcal{A}$ is the mixed strategy adopted by p when faced with type $i \in \mathcal{A} \cup \{p\}$. The symbol $\Delta\mathcal{A}$ denotes the n -dimensional simplex, such that z_{ik} is the probability that the plastic type plays strategy k against type i , which implies that $\sum_{k \in \mathcal{A}} z_{ik} = 1$. With these definitions, the Cartesian product $\prod_{i=1}^{n+1} \Delta\mathcal{A} = (\Delta\mathcal{A})^{n+1}$ is the strategy set of the plastic type. The mixed strategy $\mathbf{z}_{n+1} = \mathbf{z}_p \in \Delta\mathcal{A}$ is the strategy adopted by a plastic individual when faced with another plastic individual. Note that the existence of $\mathbf{z}_p$ does not imply that the plastic type "recognizes" itself; $\mathbf{z}_p$ could also just be seen as the default strategy of the plastic type when it cannot tell with certainty the pure type of its opponent. We explore in greater depth the consequences of an imperfect perceptual system later in this paper. The payoff of type $i \in \mathcal{A}$ against type $j \in \mathcal{A}$ is denoted $\pi(i, j)$, with the convention that the payoff goes to the individual whose strategy appears in the first position in parentheses. For interactions involving the plastic type p , we generally write the payoff $\pi(p, i) = \pi(\mathbf{z}_i, i)$ to emphasize the dependence of the payoff on the mixed strategy $\mathbf{z}_i$ of type p against i . We identify the payoffs $\pi(\mathbf{z}_i, i)$ and $\pi(i, \mathbf{z}_i)$ with the expected payoff generated by the mixed strategy of the plastic type, that is

$$\pi(\mathbf{z}_i, i) = \sum_{j \in \mathcal{A}} z_{ij} \pi(j, i) \quad \text{and} \quad \pi(i, \mathbf{z}_i) = \sum_{j \in \mathcal{A}} z_{ij} \pi(i, j). \quad (1)$$

When two plastic individuals meet, they both use their strategy $\mathbf{z}_p$ against a plastic type, so their payoff reads

$$\pi(\mathbf{z}_p, \mathbf{z}_p) = \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{A}} z_{pi} z_{pj} \pi(i, j), \quad (2)$$

where we assumed that both plastic types adopt the same responsive strategy. We do not consider selection on the plastic response itself, $\mathbf{z}$, in the analytical part of this paper, so all plastic types will always have the same responsive strategy (see at the end of this paper numerical simulations exploring the possibility that different plastic individuals may have different responses). We are interested in tracking the vector of frequencies of the types $\mathbf{x} = (x_1, \dots, x_n, x_{n+1}) \in \Delta^{n+1}$ such that $\sum_{i=1}^{n+1} x_i = 1$. Even though we restrict our study to the evolutionary competition between pure types and plasticity, we define the following expected payoffs associated with mixed strategies, which will be useful in proving results using compact notation. Namely, the expected payoff to a player adopting mixed strategy $\mathbf{x} \in \Delta\mathcal{A}$ against a player adopting mixed strategy $\mathbf{y} \in \Delta\mathcal{A}$ is defined, similarly to eq. 2, as

$$\pi(\mathbf{x}, \mathbf{y}) = \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{A}} x_i y_j \pi(i, j). \quad (3)$$

In particular, the payoff at the mixed Nash equilibrium (NE) $\mathbf{x}_{\text{NE}} \in \Delta\mathcal{A}$ of the pure-type game is written as $\pi(\mathbf{x}_{\text{NE}}, \mathbf{x}_{\text{NE}})$. We write $w_i(\mathbf{x}), i = 1, \dots, n$, for the fitness of type i when the population is in state $\mathbf{x}$ which is calculated as the average payoff at state $\mathbf{x}$, or

$$w_i(\mathbf{x}) = \sum_{j=1}^{n+1} x_j \pi(i, j). \quad (4)$$

We further assume that type p pays a cost $k > 0$ for expressing a plastic response so that its fitness reads

$$w_p(\mathbf{x}) = \sum_{j=1}^{n+1} x_j \pi(p, j) - k. \quad (5)$$

The frequency of any type i evolves according to the replicator dynamics, which are given by the differential equations

$$\dot{x}_i = x_i (w_i(\mathbf{x}) - \bar{w}(\mathbf{x})), \quad i \in \mathcal{A} \cup \{p\}, \quad (6)$$

where

$$\bar{w}(\mathbf{x}) = \sum_{i=1}^{n+1} x_i w_i(\mathbf{x}) \quad (7)$$

is the average fitness in the population at state $\mathbf{x} \in \Delta^{n+1}$. A further element of notation is that we denote by $\phi(\cdot, \mathbf{x}_0) : \mathbb{R}^+ \rightarrow \Delta^{n+1}$ the global solution trajectory (the *flow*) to eq. 6 that passes through $\mathbf{x}_0$, which means that $\phi(t, \mathbf{x}_0)$ is the vector of frequencies at time t given that the vector of frequencies passes through $\mathbf{x}_0$. We denote by $\phi_i(t, \mathbf{x}_0)$ the i^{th} element of $\phi(t, \mathbf{x}_0)$, with $i \in \mathcal{A} \cup \{p\}$.

3 2×2 games of cooperation

We begin by studying simple examples of games in order to form intuition about the evolutionary success of plasticity in social interactions. More specifically, we consider three 2×2 games of cooperation traditionally studied in evolutionary biology — the Prisoner’s Dilemma game (PD), the Stag-hunt game (SH), the Snowdrift game (SD) — as well as a simple resource consumption game. We label the two possible actions as C (for Cooperation) and D (for Defection), and this generates a plastic type with a three-dimensional trait $z = (z_C, z_D, z_p) \in [0, 1]^3$, where z_i is the probability to cooperate when facing type $i \in \{C, D, p\}$. We perform a sensitivity analysis with respect to the parameters z_C, z_D, z_p . For each value of these parameters, we determine the fate of plasticity when competing in populations consisting of pure types. We will use the standard notation for the payoffs of 2×2 social dilemma games, often used in the biological literature, i.e., $\pi(C, C) = R$, $\pi(C, D) = S$, $\pi(D, C) = T$, and $\pi(D, D) = P$ (Fig. 1).

	C	D		C	D	p
C	R	S	C	R	S	$\pi(C, z_C)$
D	T	P	D	T	P	$\pi(D, z_D)$
			p	$\pi(z_C, C)$	$\pi(z_D, D)$	$\pi(z_p, z_p)$

Fig. 1: A generic 2×2 game (left) with two available actions, C and D , and its extension with the plastic type (right) generated by the strategy $z = (z_C, z_D, z_p)$. The phenotype of the focal player is in the row and the phenotype of its opponent is in the column. Payoffs are to the row player. The payoffs to and against a plastic type are defined in eqs. 1–2.

3.1 Prisoner’s dilemma game

In this section we define the various conditions for plasticity to invade pure types or to resist invasion, using the generic payoffs R , S , T , and P , so the inequalities below are valid for any 2×2 symmetric normal-form game. However, in this particular section we discuss the validity of these inequalities for the Prisoner’s Dilemma game. The payoffs of the Prisoner’s dilemma satisfy $T > R > P > S$ and $R \geq (T + S)/2$ (even though the game is not repeated, the last condition will play an important role). In this game, there is one dominant action, D . The outcome (D, D) is the only stable equilibrium of the pure-type game. It follows that the plastic type cannot invade a population of defectors, which means that the inequality

$$\pi(i, i) < \pi(z_i, i) - k \quad (8)$$

can never be satisfied for $i = D$. But can plasticity invade cooperators, and is it immune against invasion by pure types? As to the first question, setting $i = C$ in eq. 8 we find that any strategy such that $z_C < (T - R - k)/(T - R)$ would guarantee that plastic mutants invade a monomorphic population of cooperators (i.e., the plastic type should defect with positive probability against cooperators). As to the second question regarding the ability of plastic types to resist invasion by pure types, we have to solve the inequalities

$$\pi(i, z_i) < \pi(z_p, z_p) - k \quad (9)$$

for z_i and z_p , replacing successively $i = C$ and $i = D$. In Fig. 2A–B, we show the regions of the space defined by combinations of (z_D, z_p) and (z_C, z_p) that satisfy eq. 9, which constitute the set of plastic types that are immune against the invasion by defectors and cooperators respectively in the Prisoner's dilemma game. Note that eq. 9 implies that the plastic type can be immune against the invasion by defectors only if $z_p > 0$ (this can be shown by setting $i = D$ and $z_p = 0$ in eq. 9, which then cannot be satisfied because $T > P$).

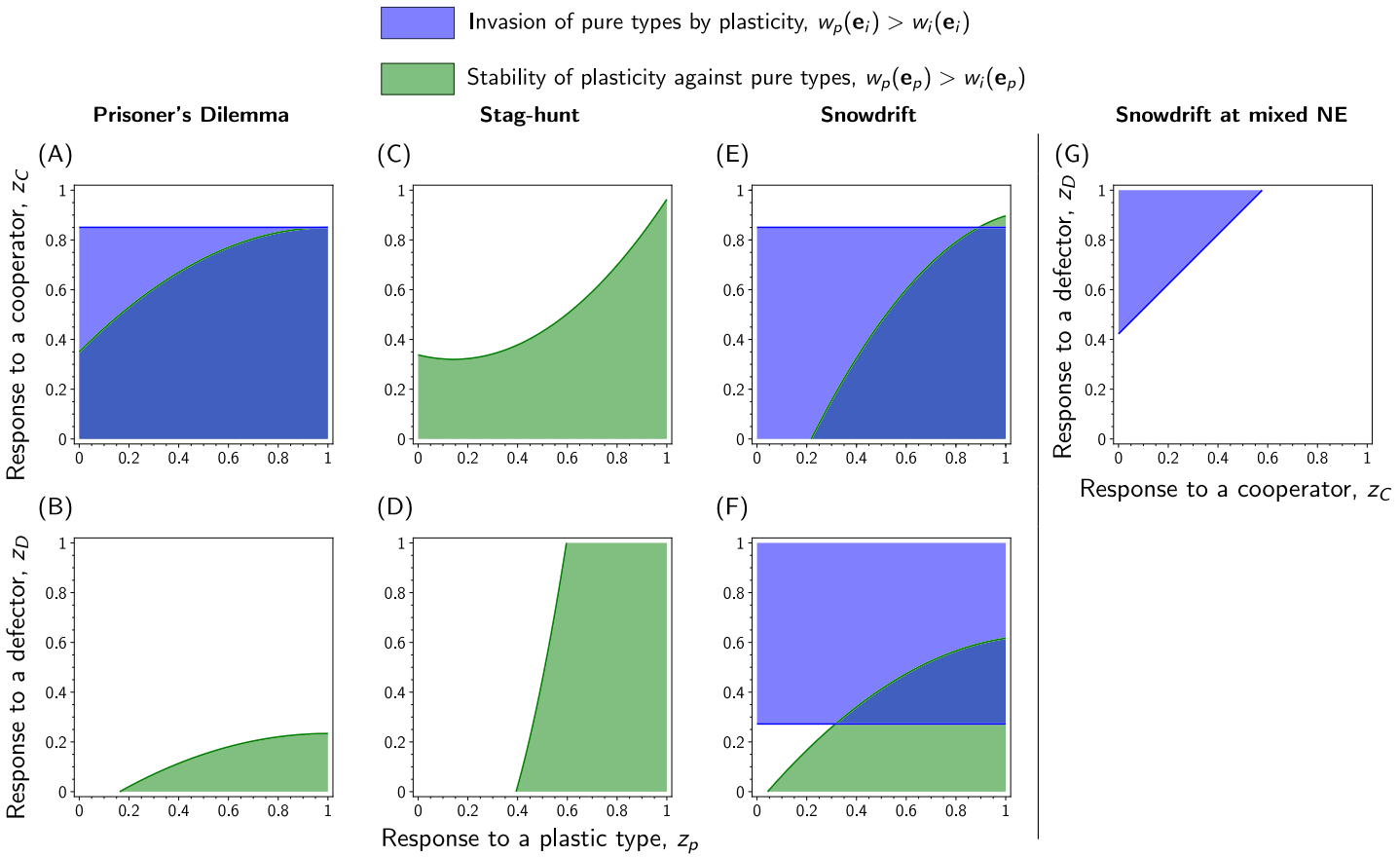


Fig. 2: Regions of stability and invasion of plastic types. (A)–(F) For each social dilemma game (columns) we show the response of the plastic type, that is the combinations (z_p, z_C) (top row) and (z_p, z_D) (bottom row) that allow invasion of monomorphic populations of pure types by plasticity (blue), and the regions that allow stability of plasticity against the invasion by pure types (green). We recall here that z_i is the probability to cooperate of the plastic type against each type $i \in \{C, D, p\}$. (G) Combinations of (z_C, z_D) that allow plasticity to invade a population of pure types at the mixed ESS of the Snowdrift game. Parameter values: Prisoner's dilemma game: $(R, S, T, P) = (3, 1, 5, 2)$; Stag-hunt game: $(R, S, T, P) = (8, 0, 4, 3)$; Snowdrift game: $(R, S, T, P) = (3, 0.1, 5, -1)$; $k = 0.3$. In the legend, $\mathbf{e}_j \in \Delta^{n+1}$ denotes the vector of type frequencies where type j is at frequency 1, while all other types are at frequency 0.

What are the conditions on the payoffs of the game, (R, S, T, P) , that allow these inequalities to be satisfied? The feasible payoffs of a plastic type as a resident (the right-hand side of eq. 9) are on the line

$x = y$ within the convex hull of feasible payoffs. This means that the payoff $\pi(z_p, z_p)$ satisfies

$$\min\{R, P, \frac{T+S}{2}\} \leq \pi(z_p, z_p) \leq \max\{R, P, \frac{T+S}{2}\}. \quad (10)$$

Combining the above inequality with eq. 9, we then see that plasticity cannot be locally stable against mutant cooperators if

$$\max\{R, P, \frac{T+S}{2}\} \leq \min\{R, S\}. \quad (11)$$

Similarly, using eq. 9, plasticity cannot be locally stable against mutant defectors if

$$\max\{R, P, \frac{T+S}{2}\} \leq \min\{T, P\}. \quad (12)$$

The last two inequalities are not satisfied in the Prisoner's dilemma, which means that a plastic type can always find a strategy that makes it stable against the invasion by pure types [EXPLAIN HERE THAT A z_p THAT RESISTS INVASION AGAINST DEFECTORS NECESSARILY RESISTS INVASION AGAINST COOPERATORS?].

In sum, in the Prisoner's Dilemma game, the best outcome for a plastic type is local asymptotic stability. It is impossible for plasticity to invade a population at the stable equilibrium of the pure-type game because the plastic type cannot do better than pure defectors.

3.2 Stag-hunt game

We define this game as one with payoffs satisfying $R > T > P > S$ and $R + S > T + P$, which entails that (C, C) and (D, D) are two pure Nash equilibria (NE) of the game, with (C, C) being the Pareto-dominant equilibrium, and (D, D) being the risk-dominant equilibrium. Consequently (C, C) and (D, D) are both asymptotically stable equilibria of the replicator dynamics in the pure-type game. Solving inequality 8 by replacing the payoffs of the Stag-hunt game, we find that plastic types cannot invade any monomorphic population of pure types. However, solving inequality 9, we find that plasticity can be locally stable (Fig. 2C–D). This game is a good illustration of the idea that invasion ability and local stability of our plastic types might be impossible to reconcile. Indeed, the plastic type needs to anti-coordinate with C and D in order to ensure local stability, but the best it can do in monomorphic populations of either C or D is to coordinate; and this best effort is not even enough to be able to invade monomorphic populations.

We finally mention a result in the Stag-hunt game that may be of interest beyond evolutionary biology. Namely, in our model, costless plasticity in the Stag-hunt game is a solution to the problem of converging to the payoff-dominant equilibrium. Even though our main focus in this paper is on the case where the cost of plasticity is positive ($k > 0$), we mention this result because most evolutionary or learning processes known to us converge to the risk-dominant equilibrium, (D, D) (or stochastic evolutionary processes admit a stationary distribution that puts more mass on the risk-dominant equilibrium, e.g., Young, 1993). Here, if one sets $k = 0$ and $(z_C, z_D, z_p) = (1, 0, 1)$, then the face of the simplex such that defectors are at frequency 0 is globally stable. All points on this face are neutrally stable, i.e., any mix of cooperators and plasticity is globally stable. In such populations, everyone cooperates and thus achieves the maximum possible payoff in the Stag-hunt game.

In sum, in the Stag-hunt game, the best outcome for costly plasticity is local asymptotic stability, which is achieved through anti-coordination with the pure types. A plastic type cannot invade monomorphic populations because the best it can do is to coordinate with them but the plastic type must pay a cost. We finally noted that if we allow plasticity to be costless, then this can make cooperation the outcome of natural selection from any initial condition, even when the population is close to the equilibrium where everyone defects.

3.3 Snowdrift game

In the Snowdrift game, the payoffs are such that $T > R > S > P$. In evolutionary biology the Hawk-Dove game also obeys this payoff ordering. This game calls for closer attention because it has a symmetric mixed NE, which is the unique ESS of the pure-type game. Thus, in order to achieve global stability, a plastic type must be able to invade the NE mixture. This game is in fact an excellent illustration of the idea that playing the best response to every pure type does not always yield the best outcome for a plastic type. Indeed, in this game, playing the best response to cooperators and defectors guarantees that plasticity will invade monomorphic populations and the mixed NE, but this strategy does not guarantee local stability of plasticity. However there exists a strategy, which is not the best response, that guarantees global stability of plasticity, provided the payoffs allow it [I SEEM TO ASSUME HERE THAT GLOBAL STABILITY OF PLASTICITY IS A NECESSARY CONSEQUENCE OF THE PLASTIC TYPE BEING ABLE TO INVADE ALL EQUILIBRIA OF THE ORIGINAL GAME AND BEING ABLE TO RESIST INVASION BY PURE TYPES; I'M NOT SURE THIS REALLY HOLDS IN THE HAWK-DOVE GAME.]. Such a strategy is found by solving simultaneously the inequality in eq. 8, together with the following inequality

$$\pi(\mathbf{x}_{\text{NE}}, \mathbf{x}_{\text{NE}}) < x_{C,\text{NE}}\pi(z_C, C) + x_{D,\text{NE}}\pi(z_D, D) - k, \quad (13)$$

which is the condition for plasticity to invade a population at the mixed NE of the Snowdrift game. It turns out to be quite difficult to reduce the inequalities in eq. 8 and eq. 13 (this is a system of mixed linear and quadratic inequalities), but we provide in Fig. 2E–G an example showing that there exists a strategy and payoffs of the Snowdrift game such that they all hold, i.e., such that plasticity is globally stable.

The Snowdrift game also provides important counter-examples to conjectures that one might be tempted to make: if a plastic type can invade every monomorphic population, it cannot necessarily invade a population at the mixed NE; in Fig. 2E–G, take for example $(z_C, z_D, z_p) = (0.2, 0.4, z_p)$. The converse is also false: if a plastic type can invade the mixed NE, this does not mean that it can invade every monomorphic population in the support of the NE. However, in any game without pure NE, there always exists a strategy that guarantees invasion of both mixed NE and pure monomorphic populations; in particular, playing the best response against every pure type always guarantees such an outcome.

Another statement that can be proved wrong using the Snowdrift game is that games without pure symmetric NE always make possible global stability of plasticity. In the Snowdrift game, setting T very large creates a game that has still no pure symmetric NE, but where there is no longer a strategy that makes plasticity globally stable. The intuitive reason is that setting T very large reduces substantially the upper bound on z_D such that plasticity is locally stable, i.e., the (green) region of local stability of plasticity in Fig. 2F shrinks and then no longer intersects the (blue) region where plasticity can invade a monomorphic population of defectors.

In sum in the Snowdrift game, we found that playing the best response allows the plastic type to guarantee co-existence with the pure types. We also found that a subset of Snowdrift games, where T is not too large, also allows the plastic type to become globally asymptotically stable.

3.4 A simple resource consumption game

In order to illustrate the range of possible outcomes for plasticity, we finally provide a game where plasticity can neither invade pure types nor be locally stable, irrespective of its strategy z . Consider the very simple situation where two individuals have access to a resource of value b . Now action C corresponds to consuming the resource and D corresponds to not consuming it. If both individuals decide to consume it, let us assume that they must share it (this captures a situation where, e.g., both individuals are of equal strength on average, so no one can monopolize the resource), and obtain a payoff of $R = b/2$. If only one of them decides to consume the resource, she gets the whole of it, $S = b$,

while the other gets $T = 0$. Finally, if neither of them consumes the resource they both get $P = 0$. In this game it is always better to consume the resource irrespective of the opponent's action, even though one would prefer to eat it alone. Consuming the resource is a dominant action, and both players consuming is thus the unique NE of this trivial game. In general we find that in games where action C is dominant and $S \geq R$, plasticity can neither invade monomorphic populations of pure types nor be locally stable against the invasion by pure types.

In sum, in the resource consumption game, if $S \geq R$, the plastic type is unable to invade or be locally stable against pure types, while if $S < R$ the best possible outcome for plasticity is local stability. In Fig. 3, we show the phase portrait of trajectories in the resource consumption game, but also for all games of cooperation studied in this section.

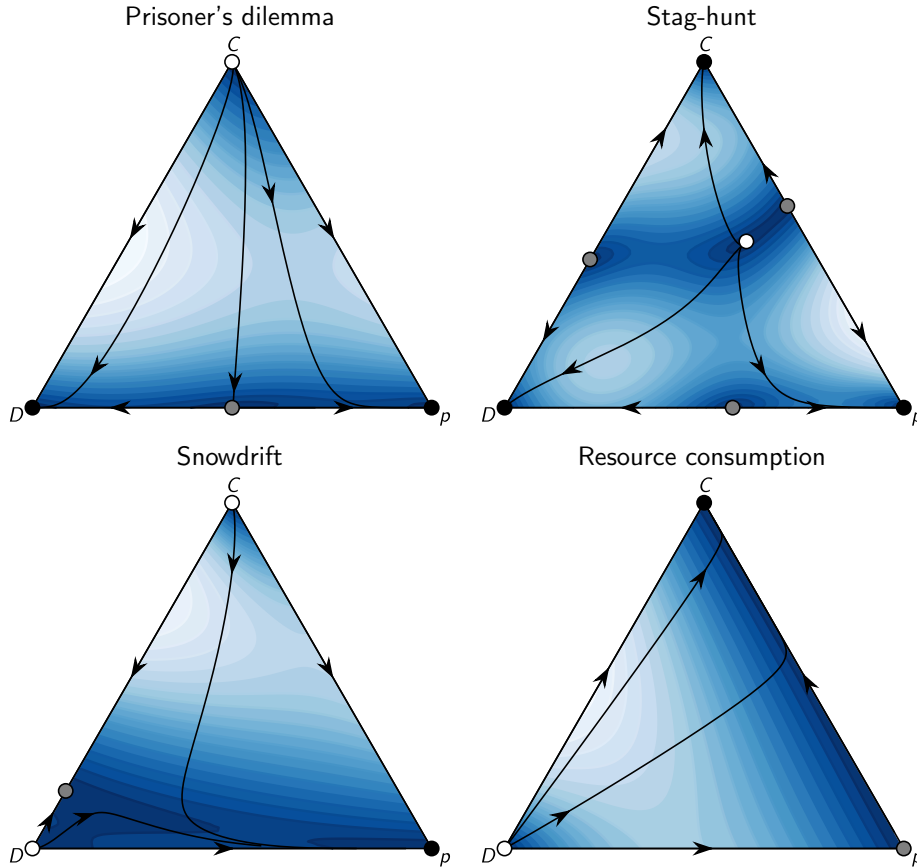


Fig. 3: Replicator dynamics with added plastic types in 4 standard 2-action, 2-player games. Each triangle represents a projection of the 3-simplex on the 2-dimensional plane, where at each vertex the corresponding strategy (C , D , or p) is at frequency 1. On the face opposed to a vertex, the corresponding strategy is absent, i.e., it is at frequency 0. These plots are displayed for the best possible outcome that can be achieved by a plastic type. Parameter values: Prisoner's Dilemma: $(R, S, T, P) = (3, -2, 5, 1)$, and $(z_C, z_D, z_p) = (0, 0, 1)$; Stag-hunt: $(R, S, T, P) = (8, 0, 4, 3)$, and $(z_C, z_D, z_p) = (0, 1, 1)$; Snowdrift: $(R, S, T, P) = (2.5, 0.1, 5, -1)$, and $(z_C, z_D, z_p) = (0.03, 0.45, 0.9)$; Mutualism: $(R, S, T, P) = (5, 6, 0, 0)$, and $(z_C, z_D, z_p) = (1, 1, 1)$; in all games we used a cost of $k = 0.3$. Contours indicate speed with darker shades corresponding to slower dynamics and lighter shades corresponding to faster dynamics. Black dot: Asymptotically stable equilibrium (a "sink") of the replicator dynamics; gray dot: an unstable saddle; white dot: an unstable source.

4 Other games

Can we generalize some of the results we derived for 2×2 games? Also, do they hold for other classes of games? In this section we first state general results regarding what is possible or impossible for plastic types and we then show that our plastic types are most favored in the special class of zero-sum games. In all of our results, we assume that the pure-type game is generic (i.e., no two outcomes yield the same

payoffs for any player).

We begin by recalling here a classical result that we will be using throughout this section. First we need to define what an Evolutionarily stable strategy is (ESS, Hofbauer and Sigmund, 1998). With our notation, an ESS is a strategy $\mathbf{x} \in \Delta^{n+1}$ such that

$$\begin{aligned} \pi(\mathbf{x}, \mathbf{x}) &\geq \pi(\mathbf{y}, \mathbf{x}), & \forall \mathbf{y} \in \Delta^{n+1}, \\ \pi(\mathbf{x}, \mathbf{y}) &> \pi(\mathbf{y}, \mathbf{y}), & \text{if } \pi(\mathbf{x}, \mathbf{x}) = \pi(\mathbf{y}, \mathbf{x}). \end{aligned} \quad (14)$$

Note that a strict NE is always an ESS.

The three statements of Prop. ?? in the Appendix have several implications and interpretations. First, statements (1) and (2) taken together imply that the task of achieving global stability is not trivial for the plastic type. If we were to take the standpoint of a plastic type that can choose its response $\mathbf{z}$ freely, then in order to maximize the chances of invading a monomorphic population consisting of type $i \in \mathcal{A}$ the plastic type would like to choose a $\mathbf{z}_i^*$ for each $i \in \mathcal{A}$ such that

$$\mathbf{z}_i^* = \arg \max_{\mathbf{z}_i} \{ \pi(\mathbf{z}_i, i) \} = \beta(i), \quad (15)$$

where we used $\beta(\cdot)$ to denote the best response function. However, statement (2) of Prop. ?? (in addition to our examples of games of cooperation studied above) shows that the best response is not the strategy that maximizes the chances of a plastic type resisting invasion by the pure types. Indeed, to achieve such stability against pure types, the plastic type should choose

$$\hat{\mathbf{z}}_i = \arg \min_{\mathbf{z}_i} \{ \pi(i, \mathbf{z}_i) \}. \quad (16)$$

The strategy $\hat{\mathbf{z}}_i$ that satisfies eq. 16 is different from the one in eq. 15, except in zero-sum games (we investigate zero-sum games in more detail below), which means that optimizing invasion ability vs. stability of plasticity can be two conflicting tasks, as we illustrated in the Stag-hunt game above. The inequality in statement (2) of Prop. ?? also implies that in order to maximize its chances of resisting invasion, the plastic type must further take the strategy $\mathbf{z}_p = \mathbf{z}_p^*$ such that

$$\mathbf{z}_p^* = \arg \max_{\mathbf{z}_p} \{ \pi(\mathbf{z}_p, \mathbf{z}_p) \}. \quad (17)$$

We have seen that in games of cooperation choosing such a $\mathbf{z}_p^*$ is equivalent to the plastic type cooperating with itself. However, if the plastic type establishes cooperation in order to be a stable strategy against pure types, one can expect that this opens the door to plastic free riders who defect against the established nice plastic types (in the Prisoner's Dilemma game and the Snowdrift game). Hence the cooperation established by our plastic types seems to be only a transient state, but to delineate the cases where this intuition holds true, it would be necessary to perform an exhaustive study of the case where the response $\mathbf{z}$ of the plastic type is under selection (we explore this question numerically in a subsequent section of this paper).

Another consequence of Prop. ?? and eq. 15 is that a plastic type cannot invade populations at a monomorphic pure NE. As we have seen in the Snowdrift game example above, this does not mean that every game without pure NE allows for the global stability of plasticity. However, we show in the next proposition that in games without pure NE, there always exists a strategy that guarantees at least co-existence of plasticity and the pure types.

We just defined the conditions for plasticity to persist in populations of pure types, but these do not guarantee local stability of plasticity. The following proposition establishes a necessary condition for the *inability* of plasticity to be immune against invasion by pure types.

The converse is not true. If (i, i) is Pareto-efficient, it might still be possible to find strategies of the plastic type such that it resists invasion by i . Taking any of the three social dilemmas above, cooperation is Pareto-efficient, but it is always possible to find a strategy that guarantees local stability of plasticity.

4.1 Zero-sum games

From the discussion of the previous sections, it seems that it is not easy to find classes of games where one can guarantee the global stability of plasticity. Looking more closely at our previous results, however, one might infer that if the two tasks of maximizing one's own payoff and minimizing the other's payoff are two compatible tasks, plastic types can achieve global stability. Hence, the class of *strictly competitive games* would seem to favor plastic types. A game is strictly competitive if, for all $\ell \in \mathcal{A}$ and any pair of strategies $i \in \mathcal{A}, j \in \mathcal{A}$, the inequality $\pi(i, j) > \pi(\ell, j)$ implies that $\pi(j, i) < \pi(j, \ell)$. In particular, maximizing one's own payoff is equivalent to minimizing the opponent's payoff. Note that in a strictly competitive game, all strategy pairs are Pareto-efficient since switching from any action that induces a higher payoff for a focal player induces a loss for his opponent. It has been shown that strictly competitive games are affine transformations of zero-sum games (Adler et al., 2009). We show in the Appendix that in the class of symmetric zero-sum games without pure NE, there always exists a strategy of the plastic type that guarantees its global stability under the replicator dynamics.

Note that the requirement that the game possesses no pure symmetric NE excludes symmetric 2×2 zero-sum games, since these games necessarily have a dominant action. Another remark is that the proof of our main result for zero-sum games is also valid for strictly competitive games since our argument is independent of affine transformations to the payoffs of the game. This remark allows us to apply our results to the *good* Rock-Paper-Scissors game, where winning yields a payoff of a , while losing induces a loss of b , and $a > b$ (Sandholm, 2011), which entails the payoff matrix

$$\begin{array}{c} \begin{array}{ccc} & R & P & S \\ \begin{array}{c} R \\ P \\ S \end{array} & \begin{bmatrix} 0 & -b & a \\ a & 0 & -b \\ -b & a & 0 \end{bmatrix} \end{array} \end{array}, \quad (18)$$

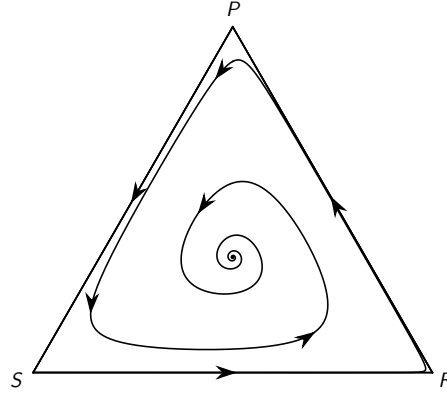
where the row and column labels R, P , and S correspond to the three possible actions (Rock, Paper, Scissors), and the payoffs are to the row player. In such a pure-type game, the interior NE, $\mathbf{x}_{\text{NE}} = (1/3, 1/3, 1/3)$, is globally stable under the replicator dynamics, with solution trajectories displaying damped oscillations around this equilibrium. In Fig. 4, we show the effect of adding a plastic type with a strategy described in the proof of Prop. 3 to the good RPS game. In this game, this creates a plastic type that plays the best response to any pure type and plays any $i \in \mathcal{A}$ against itself such that $\mathbf{z}_i = \beta(i)$ for any $i \in \{R, P, S\}$ and $\mathbf{z}_p = (1, 0, 0)$ [the choice of always playing R is arbitrary, in accordance with the proof of Prop. 3]. Even though the plastic type cannot invade on the faces of the simplex of dimension 2 (because symmetric 2×2 zero-sum games necessarily have a dominant action, and hence a pure symmetric NE), it achieves a larger payoff than any fully mixed population. This explains why it is globally stable in the interior of the simplex.

5 Model extensions

Our central goal in the previous parts of this paper was to show that even if plastic types can detect with perfect accuracy the type of their opponent and may employ any mixed strategy as a response to each type, plasticity is unable to achieve full success as long as it is costly to be plastic. However, the perfectly responsive types that we have considered are unlikely to emerge by spontaneous mutation out of a background of non-plasticity. Indeed, it is more plausible that, initially, plastic types may perform poorly in terms of type detection. Moreover, there should be natural selection on the plastic response itself, a feature we ignored above by treating the plastic response, $\mathbf{z}$, as a parameter.

In this section, we explore numerically the effect of including these more realistic assumptions in our model. We focus throughout on the Snowdrift game, because it is the game that seems to offer the richest evolutionary dynamics in our above exploration of simple 2×2 games.

"Good" Rock-Paper-Scissors game



With added plastic type

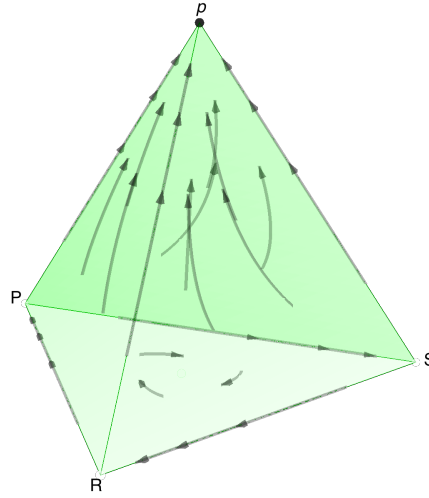


Fig. 4: Replicator dynamics in the original good RPS (top), and with added plastic type (bottom). The 3-strategy simplex was produced using our own code (available), while the 4-strategy simplex was produced using Dynamo (Franchetti and Sandholm, 2013). Parameter values: Payoff matrix of eq. 18 with $a = 2$ and $b = 1$; $\mathbf{z}_R = (0, 1, 0)$; $\mathbf{z}_P = (0, 0, 1)$; $\mathbf{z}_S = (1, 0, 0)$; $\mathbf{z}_p = (1, 0, 0)$.

5.1 Imperfect type recognition

We now examine the effect of the plastic type making errors in detecting the type of their opponent. We consider two different error probabilities: ε the probability to misidentify a pure type, and ε_p the probability to misidentify a plastic type. We capture these errors by defining the *effective mixed strategy*, $\mathbf{q}_i$, of a plastic type against pure type $i \in \mathcal{A}$ as

$$\mathbf{q}_i = (1 - \varepsilon)\mathbf{z}_i + \frac{\varepsilon}{n} \sum_{j \in \mathcal{A} \cup \{p\}} \mathbf{z}_j, \quad (19)$$

QUESTION: WHY NOT USE

$$\mathbf{q}_i = (1 - \varepsilon)\mathbf{z}_i + \frac{\varepsilon}{n-1} \sum_{\substack{j \in \mathcal{A} \cup \{p\} \\ j \neq i}} \mathbf{z}_j?$$

SINCE HAVING i IN THE SUM WOULD MEAN THAT EVEN WHEN MAKING AN "ERROR", THE RESPONSE TO i COULD STILL BE THE INTENDED ONE...

which entails that a plastic type recognizes its opponent successfully with probability $(1 - \varepsilon)$ and then applies its mixed strategy $\mathbf{z}_i$. The plastic type does not recognize the opponent with probability ε , and

in this case employs its average mixed strategy over all possible types of the opponent. The effective mixed strategy against a plastic type, $\mathbf{q}_p$, is defined similarly as

$$\mathbf{q}_p = (1 - \varepsilon_p)\mathbf{z}_p + \frac{\varepsilon_p}{n} \sum_{j \in \mathcal{A} \cup \{p\}} \mathbf{z}_j, \quad (20)$$

where ε_p is the probability to fail recognizing the plastic type as plastic.

5.2 Selection on the plastic response, $\mathbf{z}$

Assuming $\varepsilon = \varepsilon_p = 0$, we focus on the evolution of the plastic response through agent-based simulations of the Wright-Fisher process. An important question to address here is whether the responses that are optimal against pure types, which we identified in the analytical model, are also evolutionarily stable. For example, in the Hawk-Dove game, does natural selection lead to the evolution of a response such that $z_C < z_D < (R - P)/(T - P)$ and $z_p = 1$?

5.2.1 Pure greenbeards

We have seen before that plastic types benefit immensely from the capacity to adopt a specific response, $\mathbf{z}_p$, to other plastic types. What would happen to our model predictions if plastic types would only be able to distinguish between themselves vs. others? In other words, we imagine a scenario where plastic types have two traits: the response to themselves $\mathbf{z}_p$; the response to others $\mathbf{z}_o$, which they use against any pure type $i \in \mathcal{A}$. With these assumptions, our plastic types behave as greenbeards.

5.2.2 No beards

An interesting case that corresponds to the opposite of pure greenbeards is a scenario where plastic types can distinguish among the pure types but have no particular strategy against themselves. Biologically this may stem from the absence of clear phenotypic cue attached to being a plastic type. We assume here that an encounter between plastic types results in them using their average mixed strategy over all pure types, i.e.,

$$z_p = \frac{z_C + z_D}{2}, \quad (21)$$

which entails that a plastic type is mistaken for a cooperator vs. a defector half of the time.

We numerically re-derive the regions of stability and invasion in the space of all possible 2×2 games.

5.3 Relaxing all assumptions

We now consider the co-evolution of $\varepsilon, \varepsilon_p, \mathbf{z}$ in the Snowdrift game under the Wright-Fisher model. Reproduction is haploid and asexual, the population is of constant size N , and generations are non-overlapping. Every individual produces a number of offspring $W_i = w_i/\bar{w}$, where w_i is defined as in our above model with the caveat that now we do not allow for negative fitness (this is achieved by adding a constant to all individuals' fitness such that $\min_{i \in \{N\}} \{w_i\} = 0$), and $\bar{w}$ is the average absolute fitness in the population.

A simulation of the evolutionary process unfolds as follows. We start at generation $t = 0$ with a population where no plastic type is present and the pure types are at their ESS frequencies. At each generation $t \geq 0$, each individual (who is initially a pure type) may independently mutate to become a plastic type with a small probability μ_a (the mutation rate). This initial mutant has $\varepsilon = 1 - \delta$, $\varepsilon_p = 1 - \delta$, where δ is a small perturbation, and $\mathbf{z}$ is drawn randomly from a uniform distribution. Once a plastic type has appeared in the population and produces at least one offspring, this offspring is itself subject to a different mutation rate μ_p (but we generally consider $\mu_a = \mu_p$ in our simulations). This mutant plastic type adopts a response and detection-error probabilities that are small deviations from her parent's

traits, i.e., $\varepsilon \pm \delta, \varepsilon_p \pm \delta, z \pm \delta$, where the direction of the mutation is drawn independently for each trait ($\varepsilon, \varepsilon_p, z_C, z_D, z_p$), and is positive or negative with probability 1/2 each. These mutation deviations are of course subject to boundary conditions to prevent the traits from going out of the $[0, 1]$ range.

6 Discussion

In this work, we considered the introduction of plastic types in populations consisting of simple individuals adopting pure strategies in evolutionary symmetric games. We showed that despite having the capacity of responding in any possible way to other types in the population, there does not exist a general strategy for plasticity to eliminate pure types from the population. There are several explanations to this. The first one is that in games with pure symmetric Nash equilibria, costly plasticity cannot invade an equilibrium population. Moreover, we demonstrated that there is often a conflict between optimizing invasion success and optimizing stability for plastic types, the best illustration of this result being the Stag-Hunt game, where these two tasks (optimizing invasion ability versus stability) lead to totally different behavioral responses of the plastic types. We also saw that the success of plasticity is dependent on the class of games played in the population. There are games, such as zero-sum games without pure symmetric NE, where there always exists a strategy that grants plasticity global stability under the replicator dynamics. On the other hand, there are games, such as the very simple resource consumption games that we designed, where plasticity can neither invade an equilibrium population nor be stable against the invasion by pure types.

In our more detailed investigation of games of cooperation, we found that in the Prisoner's Dilemma and the Snowdrift game, plastic types should cooperate with themselves with a strictly positive probability in order to be immune against the invasion by pure types. However, invasion success of pure populations by plasticity is generally uncorrelated to playing a cooperative strategy. In the Stag-hunt game it is just impossible for costly plasticity to invade an equilibrium population. However, costless plasticity is able to do so, and can even establish cooperation (the payoff-dominant outcome) in a population that was initially at equilibrium for defection (the inferior risk-dominant outcome). This result is similar to what [Robson \(1990\)](#) found in an earlier work (we compare our work with that of [Robson, 1990](#), in more detail below).

It is insightful to compare our results to those of previous studies of the evolution of plasticity. In particular, previous work on frequency-independent selection (e.g., [Gomulkiewicz and Kirkpatrick, 1992](#)) has revealed that variance in fitness is a key factor driving the evolution of plasticity. In our setting, we notice that in games with pure Nash equilibria (NE), the population does not experience variance in payoffs at equilibrium. Hence, it makes sense that there is no advantage to adapt behavior to different circumstances, since there is only one circumstance which is to face the pure type that is evolutionarily stable. However, in games without pure NE, there is always variance in payoffs because one might be matched with any type in the support of the mixed NE of the game. This explains why plasticity can always invade a mixed NE, because it has the possibility to express different responses to the different existing types and potentially obtain a payoff that is greater than the NE payoff. Altogether, these results suggest that in general the ability to collect more fitness-relevant information is not always beneficial, as long as this information is costly to acquire (even when costs are very small), as other studies of the evolution of learning have previously shown ([Wakano et al., 2004](#); [Nakahashi et al., 2012](#); [Aoki and Feldman, 2014](#); [Dridi and Lehmann, 2015](#)).

While our results fit well within the literature on plasticity evolution theory, there is also empirical evidence that our assumptions about plasticity correspond to some extent to real-world instances of plastic phenotypes. If the evolving behavioral trait of interest is linked to morphological or hormonal characteristics, then it could be possible for a plastic type to base their strategy on, e.g., a chemical detection mechanism, vision or smell. This is the case for many traits that indicate the quality of an

individual, which will then determine the payoff of individuals interacting with it, such as size or weight in fights (Riechert, 1978; Maynard-Smith and Harper, 2004; Arnott and Elwood, 2009); this is also the case for traits that indicate the compatibility and quality of potential mates (Potts et al., 1991; Roberts and Gosling, 2003).

The ability to gather information about conspecifics or the environment is critical in evolution and, as a consequence, the proposition that individuals with more information may have an evolutionary advantage in social interactions is not new. Robson (1990) developed a model where his plastic type could only distinguish between plastic and non-plastic types and condition their action on this cue. He assumed that plasticity was costless and found in particular that plasticity can invade a population at the inferior equilibrium of a coordination game, and lead to the superior payoff-dominant equilibrium; however his plastic type could not co-exist with the superior pure type that plays the payoff-dominant action, in contrast to what we found. Banerjee and Weibull (1995) studied a very similar model to ours, but constrained the plastic type to play a best response to every pure type. We saw that this best response assumption is not the best strategy for plastic types when it comes to evolutionary stability. This phenomenon is exemplified by the Snowdrift game, where playing the best response to every pure type does not allow plasticity to be locally stable against the invasion of pure mutants; but there are Snowdrift games where playing a strategy different than the best response allows plasticity to be not only locally but globally stable. Our model is thus an extension of the ideas developed in these two earlier studies (Robson, 1990; Banerjee and Weibull, 1995).

From a broader perspective, given the recent renewed interest in the evolution of strategies in the repeated Prisoner's dilemma (Press and Dyson, 2012; Adami and Hintze, 2013), our work helps bridge the gap between these recent studies and classical work on the evolution of pure non-plastic strategies in one-shot games (Hofbauer and Sigmund, 1998). Indeed, in order to be able to express a repeated-game strategy, an organism must first have the capacity to express a plastic social phenotype. Another recent trend is the focus on environmental variation in social evolution (Ashcroft et al., 2014; Dridi and Lehmann, 2014; Weitz et al., 2016), and our study reminds that environmental variation is already embedded within social evolution through frequency-dependent payoffs (Graves and Weinreich, 2017), which is why we find that plasticity may evolve even in the absence of variation in the game structure itself.

Our model finally introduces new questions and calls for further research on the topic. Most notably, we ignored evolution of the plastic response itself, but it is natural to imagine that if one plastic type successfully invades, other plastic mutants with a different behavioral response might appear and displace the original plastic type; there is then no guarantee that the optimal plastic type will also be evolutionarily stable. Indeed, Robson (1990) showed that in the Prisoner's dilemma, if a first plastic mutant might successfully invade and establish cooperation, the introduction of further plastic mutants might lead to the loss of cooperation. A possible avenue for future research would be to see if this result applies to our setting as well, in the Prisoner's dilemma and other games.

One might argue that the ability to express mixed strategies is what gives plasticity its strategic advantage rather than the ability to be plastic. It is not the case for invasion ability because best response to pure types is always a pure strategy, so a plastic type only able to play pure strategies would not be disfavored with respect to our plastic types. However the ability to play mixed strategies is critical when looking for plastic types that are immune against the invasion by pure types.

Another extension of our work would be to consider the possibility that the game payoffs change as a function of time, and in this setting a plastic type would be able to condition its action not only on the type of the opponent but also on the type of game being played. Heller (2004) modelled a situation where some individuals could detect the state of the environment and the opponent's type in a context of a fluctuating game, but she assumed that plastic types (which she calls learning agents) only play the

best response to pure types. It would be interesting to see what would happen if we do not constrain the plastic response to be the best response, since we saw in our model that the best response does not always yield the best outcome for plasticity. These and other extensions will help us better understand the evolution of complex strategies in realistic changing environments.

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Appendix A: General results

Conditions for global stability of plasticity

We begin by recalling here a classical result that we will be using throughout this section. First we need to define what an Evolutionarily stable strategy is (ESS, Hofbauer and Sigmund, 1998). With our notation, an ESS is a strategy $\mathbf{x} \in \Delta^{n+1}$ such that

$$\begin{aligned} \pi(\mathbf{x}, \mathbf{x}) &\geq \pi(\mathbf{y}, \mathbf{x}), & \forall \mathbf{y} \in \Delta^{n+1}, \\ \pi(\mathbf{x}, \mathbf{y}) &> \pi(\mathbf{y}, \mathbf{y}), & \text{if } \pi(\mathbf{x}, \mathbf{x}) = \pi(\mathbf{y}, \mathbf{x}). \end{aligned} \quad (\text{A1})$$

Note that a strict NE is always an ESS. We state this and other more general results in the following classical theorem.

Theorem (Static equilibria and the replicator dynamics; Webb, 2007, p. 183). *Let $\mathbb{A}$ be the set of asymptotically stable equilibria of the replicator dynamics (eq. 6), $\mathbb{N}$ the set of symmetric Nash equilibria of the game, $\mathbb{N}^*$ the set of strict symmetric Nash equilibria of the game, $\mathbb{E}$ the set of Evolutionarily stable states (ESS), and $\mathbb{F}$ the set of equilibria (fixed points) of the replicator dynamics (eq. 6). We then have*

$$\mathbb{N}^* \subseteq \mathbb{E} \subseteq \mathbb{A} \subseteq \mathbb{N} \subseteq \mathbb{F}.$$

The above theorem's purpose is mainly to allow us making statements about the replicator dynamics by using easier arguments based on static payoff comparisons. Another idea that will prove useful is noticing that the addition of plasticity to a pure-type two-player game with n actions results in an *extended game* with two players and $n + 1$ strategies. We illustrate this in Fig. X for games of cooperation. Depending on the response of the plastic type $\mathbf{z}$, this extended game possesses one or several NE and may possess ESSs as well. In the next proposition, we use the NE and ESS of the extended game to delineate necessary and sufficient conditions for plasticity to be globally asymptotically stable. It turns out that one only needs to verify that the original NE of the pure-type game are no longer NE in the extended game with plasticity. Additionally, one needs of course to check that plasticity is an ESS.

Survival of plasticity

Proposition 1 (Coexistence of plasticity and pure types in games with mixed NE). *In games with no pure symmetric Nash equilibrium, there always exists a strategy $\mathbf{z} \in (\Delta\mathcal{A})^{n+1}$ such that plasticity persists in the population in the long run, i.e., there exists a $\bar{k}$ such that, for all $0 < k < \bar{k}$, any $\mathbf{x} \in \Delta\mathcal{A}$ (i.e., such that $x_p = 0$) is unstable.*

Proof. To show that any state $\mathbf{x} \in \Delta\mathcal{A}$ such that $x_p = 0$ is unstable, it suffices to show that there exists a cost $k > 0$ such that $w_p(\mathbf{x}) > \bar{w}(\mathbf{x})$, for all $\mathbf{x} \in \Delta\mathcal{A}$, where we recall that $\bar{w}(\mathbf{x})$ denotes the average fitness in the population at state $\mathbf{x} \in \Delta\mathcal{A}$. Note that the fitness of any type $i \in \mathcal{A}$ at state $\mathbf{x} \in \Delta\mathcal{A}$ is

$$w_i(\mathbf{x}) = \sum_{j \in \mathcal{A}} x_j \pi(i, j). \quad (\text{A2})$$

The fitness at state $\mathbf{x} \in \Delta\mathcal{A}$ of a mutant plastic type is

$$w_p(\mathbf{x}) = \sum_{j \in \mathcal{A}} x_j \pi(\mathbf{z}_j, j) - k. \quad (\text{A3})$$

Since the game has no pure NE, there exists for each $j \in \mathcal{A}$ and $i \in \mathcal{A}$ a strategy $\mathbf{z}_j \in \Delta\mathcal{A}$ such that $\pi(\mathbf{z}_j, j) \geq \pi(i, j)$ (moreover the inequality is strict for some pure strategy $\mathbf{z}_j \in \mathcal{A}$). Let the plastic type adopt such a strategy $\mathbf{z}$. We then have that $w_p(\mathbf{x}) > w_i(\mathbf{x})$. Since the average fitness $\bar{w}(\mathbf{x})$ is a convex combination of the $w_i(\mathbf{x})$, this implies $w_p(\mathbf{x}) > \bar{w}(\mathbf{x})$. Hence the strategy $\mathbf{z}$ described makes the mutant plastic type increase in frequency at any population state $\mathbf{x} \in \Delta\mathcal{A}$, which makes such states unstable under the replicator dynamics (eq. 6) and leads to the desired result. $\square$

Before we state the next proposition, let us define a strategy $\mathbf{s} \in \Delta\mathcal{A}$ as being the *efficient* equilibrium of a game with action set $\mathcal{A}$ if

$$\pi(\mathbf{s}, \mathbf{s}) > \pi(\mathbf{s}', \mathbf{s}'), \quad \text{for all } \mathbf{s}' \neq \mathbf{s}. \quad (\text{A4})$$

Such a strategy maximizes the payoff of a monomorphic population. In our context, $\mathbf{s}$ makes part of the strategy that a plastic type should adopt in order to maximize its payoff as a resident, i.e., the plastic type should use $\mathbf{z}_p = \mathbf{s}$.

Proposition 2 (Impossible stability of plasticity). *If there exists $i \in \mathcal{A}$ such that plasticity cannot resist invasion by i , i.e. such that $w_p(\mathbf{e}_p) < w_i(\mathbf{e}_p)$, then (i, i) is the efficient equilibrium of the game.*

Proof. If plasticity cannot resist invasion by $i \in \mathcal{A}$, this means that

$$\begin{aligned} w_p(\mathbf{e}_p) &< w_i(\mathbf{e}_p) \\ \pi(\mathbf{z}_p^s, \mathbf{z}_p^s) &\leq \pi(i, \mathbf{z}_i), \quad \forall \mathbf{z}_i \in \Delta\mathcal{A}, \end{aligned} \quad (\text{A5})$$

where $\mathbf{z}_p^s = \arg \max_{\mathbf{z}_p} \{\pi(\mathbf{z}_p, \mathbf{z}_p)\}$. By way of contradiction, suppose that (i, i) is not the efficient equilibrium of the game. Then there exists $\bar{\mathbf{z}}_p \in \Delta\mathcal{A}$ such that $\pi(\bar{\mathbf{z}}_p, \bar{\mathbf{z}}_p) > \pi(i, i)$. Hence a strategy of the plastic type with $\mathbf{z}_p = \bar{\mathbf{z}}_p$ and $\mathbf{z}_i = i$ would guarantee local stability of plasticity. But, by eq. A5, this contradicts our assumption that plasticity is not locally stable against the invasion by i . This proves the result. $\square$

Zero-sum games

Proposition 3 (Global stability of costly plasticity in zero-sum games). *In zero-sum games with no pure symmetric NE, there always exists a strategy $\mathbf{z} \in (\Delta\mathcal{A})^{n+1}$ of the plastic type such that it is globally asymptotically stable under the replicator dynamics.*

Proof. In order to show that plasticity is globally asymptotically stable, we show a stronger result, i.e., that at any state $\mathbf{x} \in \text{int}(\Delta^{n+1})$, we have $w_p(\mathbf{x}) > \bar{w}(\mathbf{x})$. This implies that $\dot{x}_p > 0$ at any state $\mathbf{x} \in \text{int}(\Delta^{n+1})$ and since the simplex is a bounded set, this in turn implies that $\lim_{t \rightarrow \infty} \phi_p(t, \mathbf{x}_0) = 1$ for all $\mathbf{x}_0 \in \text{int}(\Delta^{n+1})$.

Now, to show that $w_p(\mathbf{x}) > \bar{w}(\mathbf{x})$ at all $\mathbf{x} \in \text{int}(\Delta^{n+1})$ we will show that $w_p(\mathbf{x}) > w_i(\mathbf{x})$ for all $i \in \mathcal{A}$. To do so, we write the fitness of a pure type i as

$$w_i(\mathbf{x}) = x_i \pi(i, i) + x_p \pi(i, \mathbf{z}_i) + \sum_{\substack{j \neq i \\ j \neq p}} x_j \pi(i, j). \quad (\text{A6})$$

Similarly, we write the fitness of the plastic type as

$$w_p(\mathbf{x}) = x_i \pi(\mathbf{z}_i, i) + x_p \pi(\mathbf{z}_p, \mathbf{z}_p) + \sum_{\substack{j \neq i \\ j \neq p}} x_j \pi(\mathbf{z}_j, j) - k. \quad (\text{A7})$$

In order to compare the part of the fitness that is under the summation symbol, note that since there is no pure NE, there exists for each $j \in \mathcal{A}$ a strategy $\mathbf{z}_j$ of the plastic type such that $\pi(\mathbf{z}_j, j) \geq \pi(i, j)$. Hence there is a strategy $\mathbf{z}$ such that the term under the summation symbol is larger for the plastic type than the pure type, or

$$\sum_{\substack{j \neq i \\ j \neq p}} x_j \pi(\mathbf{z}_j, j) > \sum_{\substack{j \neq i \\ j \neq p}} x_j \pi(i, j). \quad (\text{A8})$$

The remaining terms in the fitness are also larger for the plastic type than for type $i \in \mathcal{A}$. Indeed, we have $\pi(\mathbf{z}_p, \mathbf{z}_p) = 0$ for any $\mathbf{z}_p \in \Delta\mathcal{A}$. At the same time since the game is zero-sum with no pure NE, the $\mathbf{z}_i$'s chosen above to satisfy eq. A8 can simultaneously be chosen such that $\pi(i, \mathbf{z}_i) < 0$, thus

$$\pi(\mathbf{z}_p, \mathbf{z}_p) > \pi(i, \mathbf{z}_i).$$

Also, since $\pi(i, \mathbf{z}_i) = -\pi(\mathbf{z}_i, i)$, and $\pi(i, i) = 0$, we also have

$$\pi(\mathbf{z}_i, i) > \pi(i, i).$$

663 Taken together the last three inequalities imply that there exists a k such that $w_p(\mathbf{x}) > w_i(\mathbf{x})$ for all $i \in \mathcal{A}$
 664 at any state $\mathbf{x} \in \text{int}(\Delta^{n+1})$, hence $w_p(\mathbf{x}) > \bar{w}(\mathbf{x})$, which completes the proof. $\square$

	A	B	C
A	0	3	X
B	2	2	1
C	-1	$\frac{5}{2}$	3

	A	B	C	p
A	0	3	X	3
B	2	2	1	1
C	-1	$\frac{5}{2}$	3	-1
p	2	$\frac{5}{2}$	X	$f(\mathbf{z}_p)$

Fig. A1: A 3×3 symmetric game that is not strictly competitive, but where plasticity can be globally stable for sufficiently high X (left), and the extended game with added plastic type (right). In the left payoff matrix, in blue, we highlight the response of the plastic type (row) to each possible action played by the column player. The payoff of a plastic type against itself is a function of the parameter X and the strategy $\mathbf{z}_p$.

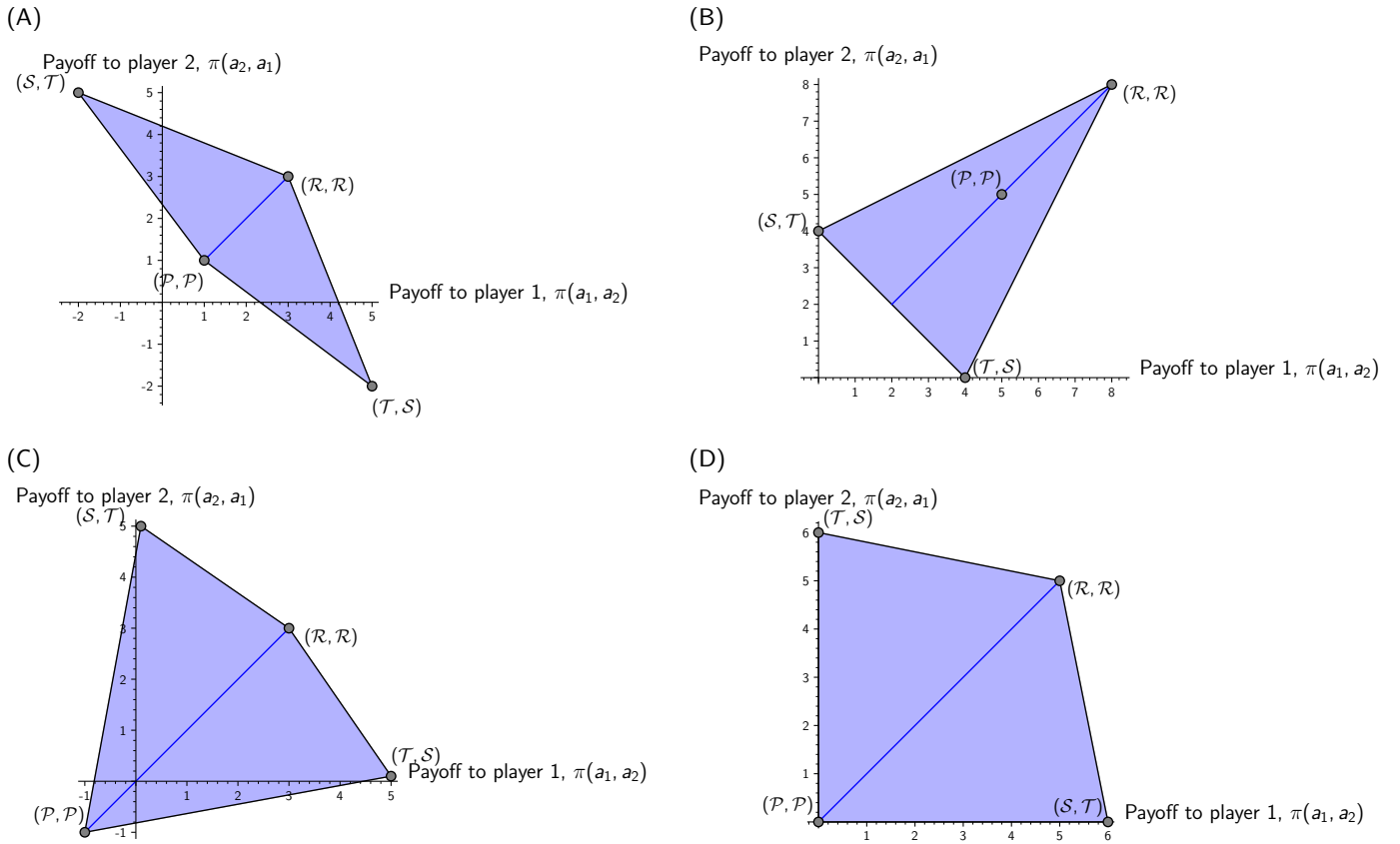


Fig. A2: The convex hull of feasible payoffs (light blue shaded area) in games of cooperation, and the line (dark blue) of feasible payoffs in a monomorphic population of plastic types. (A) Prisoner's Dilemma game with $(R, S, T, P) = (3, -2, 5, 1)$. (B) Stag-hunt game with $(R, S, T, P) = (8, 0, 4, 5)$. (C) Snowdrift game with $(R, S, T, P) = (3, 0, 1, 5, -1)$. (D). Mutualism game with $(R, S, T, P) = (5, 6, 0, 0)$.

Tab. A1: Best outcomes for plasticity in all 12 possible 2×2 game classes. For each game class (I to XII; see Fig.X) and each task in the columns, we find the best strategy $z = (z_C, z_D, z_p)$ that could be used by a plastic type; note that here we do not search for strategies that achieve all tasks *at the same time*. Thus, the strategy that allows e.g., p to invade C might be different than the strategy that allow p to be stable against invasion by C .

Game class	NEs	ESSs	p stable against C ?	p stable against D ?	p invades C ?	p invades D ?	p invades NE?
I	$\left\{ \left(\frac{P-S}{P+R-S-T}, \frac{P-S}{P+R-S-T} \right) \right\}$		✓	✗	a	a	
II			✓	✗	a	a	
III			✓	✗	a	a	
IV			✓	✗	a	a	
V			✓	✗	a	a	
VI			✓	✗	a	a	
VII			✓	✗	a	a	
VIII			✓	✗	a	a	
IX			✓	✗	a	a	
X			✓	✗	a	a	
XI			✓	✗	a	a	
XII			✓	✓	a	a	NA